

UniVoice: Breaking Communication Barriers Through AI-Powered Multimodal Messaging for Users with Sensory Impairments

Gleiston Guerrero-Ulloa, Jhon B. Leturne-Pluas, Miguel J. Hornos, and Carlos Rodríguez-Domínguez

Abstract—The digital exclusion of people with sensory disabilities from mobile messaging platforms represents one of the most pressing challenges in human-centred software engineering. This article presents the design, implementation, and empirical evaluation of **UniVoice**, an AI-powered inclusive mobile messaging system conceived for users with visual, hearing, and speech disabilities. For users with visual impairment, the system provides text-to-speech (TTS), voice command navigation, automatic image description (OCR+BLIP captioning), and audio transcription. For users with hearing impairment, the system provides speech-to-text (STT) transcription of received audio and video. For users with speech disability—which clinically decomposes into the deaf-mute subgroup and the speech-only-impairment subgroup—coverage is established through the receiver-adaptation logic of **UniVoice** without a dedicated AAC module: deaf-mute users are served by the union of the hearing-impairment adaptations (STT of received audio) and text-based composition for sending, and speech-only-impaired users retain full reception of every modality while composing messages via the mobile touchscreen, with recipient-dependent TTS when the addressee has a visual impairment. The proposed architecture integrates an intelligent processing API that orchestrates TTS, STT, automatic image description via computer vision, and optical character recognition (OCR) modules, dynamically adapting each message format to the receiver's sensory profile. The technology stack combines React Native for a cross-platform mobile interface, Spring Boot (Java) for business logic, Django (Python) for AI processing, RabbitMQ as a message broker, and PostgreSQL for persistence, with end-to-end asymmetric-key encryption. Development followed the TDDM4IoTS methodology (*Test-Driven Development Methodology for IoT-Based Systems*). **UniVoice** was evaluated with $n=25$ participants (60 % with visual impairment, 20 % with hearing impairment, and 20 % without disability) applying the Technology Acceptance Model (TAM) and a WCAG 2.1 checklist. Results revealed a global TAM mean of $\bar{x}=4.20/5$ ($\bar{x}_{PU}=4.55$; $\bar{x}_{PEOU}=3.85$), strong internal consistency ($\alpha=0.866$; PU: $\alpha=0.867$; PEOU: $\alpha=0.898$), and the highest-rated item reaching 84 % maximum-scale responses. An iterative accessibility improvement cycle produced a 64 % reduction in WCAG 2.1 warnings. The findings validate that strategic AI integration in mobile applications constitutes a viable approach to bridging the

communicative gap of people with sensory disabilities.

Index Terms—Accessibility, assistive technology, end-to-end encryption, human-machine interaction, image captioning, inclusive messaging, mobile application, React Native, sensory disability, speech recognition, speech synthesis, TAM, TDDM4IoTS, UniVoice, WCAG.

I. INTRODUCTION

INSTANTANEOUS communication has become a fundamental right in contemporary society and a cornerstone of full civic participation. Despite mobile messaging platforms surpassing three billion active users globally, their proliferation has not been matched by intrinsic accessibility architectures that guarantee equitable use for individuals with sensory impairments.

The scale of the population affected is global. The World Health Organization (WHO) estimates that approximately 15 % of the world population lives with some form of disability [1], a proportion of which concerns the three sensory profiles targeted by this work. For hearing impairment, the WHO reports that over 5 % of the world's population—some 430 million people, projected to exceed 700 million by 2050—require rehabilitation for disabling hearing loss [2]. For visual impairment, at least 2.2 billion people live with near- or distance-vision impairment worldwide, of whom at least one billion face a condition that could have been prevented or remains unaddressed [3]. For speech, voice, and language disorders, population data from national surveillance systems range from 7.7 % prevalence among U.S. children aged 3–17 [4] to 40 million Americans with a communication disorder of some form [5]. At the regional level, the Pan American Health Organization estimates that 217 million people in the Region of the Americas live with hearing loss (21.52 % of the regional population), with a projected rise to 322 million by 2050 [6]. Within this regional frame, Ecuador's National Council for Disability Equality (CONADIS) registers 124,801 individuals with active sensory disabilities, distributed as 50.30 % hearing, 45.39 % visual, and 4.31 % speech impairments, with 70,429 men (56.4 %) and 54,372 women (43.6 %) [7]. The Ecuadorian figures illustrate at a national scale the structural patterns—hearing dominant over visual, speech a smaller but non-negligible tail—consistent with the global and regional estimates.

The most widely used messaging platforms—WhatsApp, Telegram, Messenger, and Instagram—offer only partial accessibility features: limited screen-reader compatibility (Talk-

Manuscript received XX Month 2026; revised XX Month 2026; accepted XX Month 2026. Date of publication XX Month 2026; date of current version XX Month 2026. This work was supported by institutional resources of the Universidad Técnica Estatal de Quevedo (UTEQ) and the Universidad de Granada (UGR). (Corresponding author: Gleiston Guerrero-Ulloa.)

G. Guerrero-Ulloa and J. B. Leturne-Pluas are with the Faculty of Computer Sciences, Universidad Técnica Estatal de Quevedo, Quevedo 120501, Los Ríos, Ecuador (e-mail: gguerrero@uteq.edu.ec; jleturnep@uteq.edu.ec).

M. J. Hornos and C. Rodríguez-Domínguez are with the Department of Software Engineering, Universidad de Granada, 18071 Granada, Spain (e-mail: mhornos@ugr.es; carlosrodriguez@ugr.es).

Color versions of one or more figures in this article are available at <https://doi.org/10.1109/IJOT.2026.xxxxxxx>.

Digital Object Identifier 10.1109/IJOT.2026.xxxxxxx

Back and VoiceOver), contrast adjustment, and, in the case of WhatsApp, recently introduced audio-to-text transcription. However, none of these platforms comprehensively addresses the needs of users with multiple sensory profiles: bidirectional adaptation among text, audio, and image modalities; automatic semantic description of visual content; complete navigation by voice commands; nor end-to-end encryption for accessible multimodal data flows [8], [9].

A third population—users with speech (verbal production) disabilities—requires a dedicated functional analysis because “speech disability” is not a homogeneous clinical category. Standard clinical taxonomies [10]–[12] distinguish two functionally distinct subgroups on the basis of the auditory-channel status. The first subgroup comprises *deaf-mute users* (prelingual profound hearing loss in which oral speech has not developed or is not functional for communication), whose auditory and speech channels are simultaneously impaired [13], [14]. The second subgroup comprises users with *speech-only impairment*—dysarthria, aphasia, selective mutism, postlaryngectomy aphonia, or severe stammering—whose hearing and vision remain intact and whose only functional barrier is the production of voice.

Within the receiver-adaptation logic of UniVoice—the system proposed in this article—both subgroups are fully covered *without* a dedicated AAC module. Deaf-mute users share the functional barriers of the hearing-impairment profile (inability to consume incoming audio without STT mediation) and those of the speech-impairment profile (inability to produce outgoing audio without text input); the hearing-impairment adaptations already implemented in UniVoice—STT transcription of received audio/video and text-based composition of outgoing messages—therefore fully cover this subgroup with no additional engineering. Users with speech-only impairment retain unimpeded reception of every message modality (text, audio, image, video), so their reception profile is identical to that of a non-disabled user; for sending, the mobile touchscreen provides text-based composition as a universally available, fully effective alternative to voice input, and when the recipient has a visual impairment, UniVoice applies TTS to the outgoing text to preserve communicative equivalence with a voice-capable sender. This two-subgroup argument establishes that the taxonomy of the speech-disability population does not introduce a third functional case beyond those already addressed by the hearing and non-disabled reception profiles, and justifies the absence of a dedicated AAC module in the present architecture. Enhanced input assistance for severe speech disorders [11], [12]—personalised disfluency-tolerant STT, phrase banks, predictive text tailored to residual articulatory patterns—remains a complementary future-work priority rather than a gap in the current coverage claim.

From a Human-Machine Interaction (HMI) perspective, universal design [15] and the Web Content Accessibility Guidelines (WCAG 2.1) [16] require that systems be perceivable, operable, understandable, and robust for the widest possible range of users. In the mobile context, systematic accessibility evaluations of Android and iOS applications reveal non-compliance rates exceeding 60 % in criteria such as text alternatives for non-text content and complete navigation

without precise touch interaction [17], [18]. Against this background, recent advances in Artificial Intelligence (AI) offer unprecedented opportunities to overcome these barriers: state-of-the-art deep neural network TTS systems produce naturalistic speech synthesis [19], transformer-based speech-to-text models have achieved word error rates (WER) comparable to human transcription [20], and image captioning models have demonstrated practical utility for users with visual disabilities in real-world scenarios [21].

The present work is motivated by the following research questions:

- RQ1** Can a single mobile platform coherently integrate TTS, STT, image captioning, and OCR modules—together with asymmetric end-to-end encryption—while maintaining zero processing error rates under variable input conditions?
- RQ2** Does the proposed system achieve a Technology Acceptance Model score above the positive acceptance threshold of $\bar{x} > 4.0$ in a heterogeneous population of individuals with sensory disabilities?
- RQ3** Can the TDDM4IoTS development methodology, applied with accessibility as a formal acceptance criterion, produce a measurable and statistically meaningful reduction in WCAG 2.1 warnings compared to leading commercial messaging platforms?

The contributions of this work are: (i) an accessibility-oriented microservices architecture that integrates TTS, STT, OCR, image description, and asymmetric end-to-end encryption in a cross-platform mobile messaging system (addressing RQ1); (ii) a systematic application of the TDDM4IoTS methodology [22] to inclusive messaging software development; (iii) empirical evidence on technological acceptance—measured via TAM—and technical accessibility from a study involving 25 participants with heterogeneous sensory disabilities (addressing RQ2); and (iv) quantification of the impact of an iterative accessibility improvement process, achieving a global 64 % reduction in warnings identified by Google’s Accessibility Scanner (addressing RQ3).

The remainder of this article is organised as follows. Section II reviews related work along three dimensions: accessibility in mobile messaging, augmentative and alternative communication (AAC), and AI for accessibility, identifying the gaps that motivate this research. Section III describes the mixed-methods research paradigm, the eleven TDDM4IoTS phases applied—including the microservices architecture, intelligent message-conversion flow, voice-command module, asymmetric encryption, and hybrid infrastructure—and the 18 WCAG 2.1 accessibility criteria employed. Section IV presents technical API performance results (TTS, STT, image description, video), iterative accessibility results, and TAM questionnaire results. Section V interprets the findings in the context of existing literature and reflects on the implications for the Sustainable Development Goals. Section VI synthesises the main conclusions and directions for future research.

II. RELATED WORK

A. Accessibility in Mobile Messaging Applications

1) *Empirical evaluations with real users:* Weichbroth [9] conducted a systematic review of 57 usability studies on mobile applications, concluding that accessibility is among the least addressed criteria in conventional design processes. Chen *et al.* [17] carried out a large-scale empirical study of Android app accessibility—including popular messaging platforms—revealing non-compliance rates exceeding 60 % in fundamental WCAG 2.1 criteria such as text alternatives for non-text content, semantic labelling, and screen-reader compatibility.

Da Silva *et al.* [23] performed the first systematic evaluation of WhatsApp from the perspective of visually impaired users, identifying 18 critical barriers including absent alternative texts, insufficient auditory feedback on notifications, and low visibility of interactive control states. Zahida *et al.* [24] extended this approach to elderly users, documenting that absent persistent font-size adjustment, icon density on small screens, and insufficient haptic feedback are primary error sources. Swati *et al.* [25] found that Facebook Messenger is systematically inaccessible for users with complete blindness due to missing accessibility labels on critical buttons. Rumini and Astrid [26] identified that aesthetic and consistency dimensions present the greatest violations in Telegram according to Nielsen's heuristics. Kim *et al.* [27] found significant deficiencies in the emotional and affective interface design of mobile communication apps for hearing-impaired users.

2) *Systematic reviews and accessibility frameworks:* Bal-lantyne *et al.* [28] compared W3C, Google, Apple, and other normative accessibility guidelines, concluding that cross-platform fragmentation creates inconsistencies and recommending WCAG 2.1 as the transversal reference. Acosta-Vargas *et al.* [18] demonstrated that a combined automated-plus-manual approach increases detection coverage by more than 35 %. Huang *et al.* [29] evaluated the four most widely used messaging platforms with adults with mild-to-moderate cognitive disabilities, documenting systematic failures in touch-target size, colour contrast, and screen orientation. Bercaru and Popescu [8] conducted a PRISMA review of 2,819 articles concluding that adaptive personalisation of user experience remains the most deficient dimension in accessibility research.

B. Augmentative and Alternative Communication Systems

1) *Conceptual framework and base technologies:* El-sahar *et al.* [10] systematised modalities of AAC high-technology platforms and concluded that machine learning represents the most promising avenue, particularly through speech generation, text prediction, and biological signal classification. Guerrero-Ulloa *et al.* [30] presented BellChat, the first inclusive web messaging application integrating simultaneous TTS and STT for users with visual and hearing disabilities, though limited to a web-only environment. Yeratziotis *et al.* [31] integrated sign language into social media apps for deaf users; however, their approach does not address visual disability or speech impairment. Chang *et al.* [32] proposed a

smart city system combining gesture and voice recognition for hearing- and visually-impaired citizens. Janaka *et al.* [33] explored ubiquitous messaging via smart glasses, demonstrating that voice input reduces response time by 33.1 % and increases typing speed by 40.3 %.

2) *AI-based AAC applications for speech disorders:* Costanzo *et al.* [12] evaluated Talkitt—an ML application that translates unintelligible sounds into clear words—with children with Down syndrome, showing recognition precision improved from 75 % to 90 % after personalised calibration. Mateos-Sanchez *et al.* [34] developed CapacitaBOT, an IBM Watson conversational chatbot for people with intellectual disabilities. Bleakley *et al.* [11] documented critical barriers in commercial STT systems for people who stammer: excessively short end-of-utterance detection windows and inability to manage disfluencies, with direct implications for inclusive STT module design.

C. Artificial Intelligence for Accessibility

Chemnad and Othman [35] conducted a PRISMA bibliometric review of 3,706 articles (2018–2023), presenting a classification framework for AI in accessibility and concluding that research is concentrated asymmetrically on visual disabilities while hearing, speech, cognitive, and motor disabilities receive significantly less coverage. Ji *et al.* [36] and Tan *et al.* [19] established the state of the art in neural TTS, while Baeviski *et al.* [20] and Radford *et al.* [37] demonstrated WER comparable to human transcription in STT systems. Ming *et al.* [38] published the most comprehensive review of automatic image captioning to date. Dognin *et al.* [21] demonstrated on the VizWiz dataset—composed of images taken by blind people in real-world conditions—that modern captioning systems can reach utility levels comparable to human descriptions in accessibility contexts. Mukherjee *et al.* [39] showed OCR accuracy exceeding 95 % for standard printed characters. Saleem *et al.* [13], Papatsimouli *et al.* [40], and Kothadiya *et al.* [14] advanced real-time sign language recognition systems.

D. Identified Gaps

The critical analysis of the 35 reviewed works identifies five structural gaps that justify the contribution of this work.

Gap 1—Absence of multimodal mobile platforms simultaneously covering visual, hearing, and speech profiles. No reviewed work implements a mobile messaging system that dynamically adapts each message to the receiver's sensory profile combining TTS, STT, image captioning, and OCR in a single native mobile platform.

Gap 2—Technological fragmentation: mature AI components exist in isolation but have not been integrated in a coherent system with guaranteed privacy. No work combines TTS, STT, captioning, and OCR in a messaging architecture with asymmetric end-to-end encryption.

Gap 3—Scarcity of empirical TAM evidence with heterogeneous multi-disability populations. Existing studies are restricted to single disability profiles.

Gap 4—Absence of systematic, test-driven development methodologies for accessible messaging software. TDDM4IoTS has not previously been adapted to inclusive mobile messaging development.

Gap 5—Insufficient coverage of speech disorder profiles in voice-input messaging systems. No existing system offers automatic input-mode adaptation according to the user’s individual disfluency profile.

This work directly addresses all five identified gaps.

III. MATERIALS AND METHODS

This section describes, first, the materials employed—platforms, instruments, and AI components—and, second, the methods applied for system development and for technical and user evaluation. The study adopted a mixed-methods research paradigm [41]: quantitative for API conversion performance metrics (mean latency, error rate) and for the TAM questionnaire; qualitative for the analysis of accessibility barriers identified during direct user observation sessions. The research type is applied and projective [41], oriented toward the design, implementation, and evaluation of a software artifact that solves a specific technical and social problem in inclusive communication.

A. Materials

1) *Hardware Infrastructure and Deployment*: The system was deployed on a hybrid infrastructure consisting of two complementary nodes. The **AI node** is an Amazon Web Services (AWS) instance running Windows Server, assigned to the AI API (Django/Python) given its greater CPU and RAM demands for vision and language model inference. The **backend node** is a CentOS server administered by UTEQ, hosting the REST API (Spring Boot/Java), the RabbitMQ message broker, and the PostgreSQL database. Nginx served as a web server and reverse proxy on both nodes; Docker containerization ensured portability and service isolation between development and production environments. Technical API tests were executed from an Android mobile device connected to a standard WiFi network (802.11n, −55 dBm).

2) *Technology Stack*: Table I summarises the software components, selected technology, and technical justification for each choice. React Native was selected for the mobile interface for its ability to generate native UI components for Android and iOS from a single codebase, reducing maintenance costs and ensuring compatibility with platform accessibility services (TalkBack on Android, VoiceOver on iOS).

3) *Artificial Intelligence Components*: The accessible conversion modules integrate the following AI components. For **TTS**, two engines were integrated in parallel: EdgeTTS (Microsoft Neural TTS, multilingual, cloud-based) and SAPI5 (Microsoft Speech API 5, local processing via pyttsx3). For **STT**, a deep-learning transformer-based automatic speech recognition API [20], [37] was used (Whisper, executed on the AWS node). For **image description**, pre-trained vision-language models were integrated via API calls, specifically blip-image-captioning-base from Hugging Face, consistent with the state of the art [21], [42]. For

Table I: Technology Stack of the Inclusive Messaging System

Component	Technology	Justification
Mobile UI	React Native	Cross-platform; native TalkBack/VoiceOver support
REST backend	Spring Boot	Microservices, ORM, horizontal scalability
AI API	Django (Python)	ML/NLP ecosystem, pre-trained models
Real-time msg.	WebSocket + RabbitMQ	Ordered delivery, fault tolerance
Persistence	PostgreSQL	Relational, ACID-compliant, open source
Containers	Docker + Nginx	Portability, isolation, reverse proxy
Cloud (AI)	AWS Windows Server	CPU/RAM for model inference
Encryption	Asymmetric keys	End-to-end confidentiality

Table II: STRIDE Threat Analysis of Principal Communication Flows. F1 = registration; F2 = client-to-server send; F3 = server-side AI conversion; F4 = server-to-receiver delivery; F5 = key repository management. ✓ = mitigated; △ = partial; × = residual gap.

Threat	Flow/asset	Scenario	Gap	St.
Spoofing	F1: Registration	Fake account, no identity check	No 2FA documented	△
Tampering	F2/F4: Web-Socket	Message modification in transit	No independent message signature	△
Repudiation	F2: Sent msgs	Denial of authorship	No crypto non-repudiation	△
Info. Discl.	F3: Conversion	Plaintext in server memory	No TEE or memory-scrub	×
DoS	F3: Django API	Endpoint saturation	No rate limiting documented	△
Elevation	F2: REST API	Privilege escalation	No RBAC analysis documented	×

OCR, computer vision models with accuracy exceeding 95 % on standard printed text [39] were used. For **voice command navigation**, cosine similarity [43] was used as a semantic proximity metric, with cascade correction via the Llama 1B language model and a phonetic dictionary.

4) *Security and Privacy Threat Analysis*: The security architecture was analysed using the STRIDE threat modelling framework [44], applied to the five principal communication flows documented in Fig. 2 and Section III-B2. Table II summarises threats, current mitigations, and residual gaps identified for each flow.

The most critical finding concerns the **Information Disclosure** threat at the server-side conversion step (F3): the Spring Boot backend decrypts the message before forwarding it to Django, creating a transient plaintext window in server memory. The manuscript asserts that the server never stores plain text; however, no technical mechanism (memory-scrubbing, trusted execution environment, or audit log) is documented to verify this assertion. This constitutes a transparency gap; its remediation via post-processing memory-scrubbing is designated as future work.

5) *Sensory Profile Coverage Model*: Table III presents the functional coverage matrix of UniVoice across four sensory profiles, with the speech-disability row resolved according to the two-subgroup taxonomy established in Section I. Users with *speech-only impairment* retain intact auditory and visual function: reception coverage is therefore 100 % (audio messages are played directly, images are viewed directly, video is reproduced directly, and text is read visually), identical to the non-disabled reception profile. The sending channel relies on text-based composition as the primary input; when the recipient has a visual impairment, the system applies TTS conversion to the outgoing text, producing an audio message functionally equivalent to a voice-capable sender's output. Users with *deaf-mute* profile (pre-lingual profound hearing loss with non-functional oral speech) are covered through the union of the hearing-impairment column and the sending-channel adaptations of the speech-impairment column: STT transcription of received audio/video addresses the reception barrier, and text composition with recipient-dependent TTS addresses the sending barrier. This design principle—*receiver-side adaptation compensates for sender-side constraints when sensory profiles are complementary*—is revisited in Section V-D as a generalisable contribution of this work. Triple disability combinations (deaf-blind, deaf-mute-blind) are explicitly excluded: they require tactile or braille output outside the scope of a touchscreen-based mobile interface.

Table III: Functional Coverage Matrix by Sensory Profile. $\checkmark$ = fully supported; P=primary channel for this profile; $\circ$ = not applicable (requires impaired capability); $\times$ = outside scope.

Function	Visual	Hearing	Speech	None
Send text	Voz→STT $\checkmark$	Direct $\checkmark$	Text (P) $\checkmark$	Both $\checkmark$
Send audio	Direct $\checkmark$	$\circ$	$\circ$	Direct $\checkmark$
Nav. UI	Voice $\checkmark$	Touch $\checkmark$	Touch $\checkmark$	Touch $\checkmark$
Recv text	TTS $\checkmark$	Visual $\checkmark$	Visual $\checkmark$	Visual $\checkmark$
Recv audio	Play $\checkmark$	STT text $\checkmark$	Play $\checkmark$	Play $\checkmark$
Recv image	BLIP+OCR $\checkmark$	Visual $\checkmark$	Visual $\checkmark$	Visual $\checkmark$
Recv video	Caption $\checkmark$	STT subs $\checkmark$	Play $\checkmark$	Play $\checkmark$
Deaf-mute ^a	Covered by Hearing column (reception) $\cup$ Speech column (sending)			

^a Pre-lingual profound hearing loss with non-functional oral speech. Functionally equivalent to the intersection of the hearing-impairment and speech-only-impairment profiles; no additional adaptation module required.

6) *Accessibility Evaluation Framework*: The accessibility evaluation framework was built from three complementary normative sources: WCAG 2.1 [16] at conformance levels A and AA, W3C mobile accessibility guidelines, and Android/iOS guidelines systematised by Ballantyne *et al.* [28] and Acosta-Vargas *et al.* [18]. From these sources, 18 verifiable criteria were defined, organised according to the WCAG POUR principles: (i) **Perceivability**: transcription of audio and video messages, text alternatives for images, configurable synthesised speech rate, and minimum colour contrast of 4.5:1 for normal text (WCAG criterion 1.4.3); (ii) **Operability**: minimum touch target size of 48×48 dp, adaptable screen orientation, complete voice command navigation; (iii) **Understandability**: informative error suggestions, explicitly declared interface language, icon and label consistency; (iv) **Robustness**: verified compatibility with TalkBack and VoiceOver, and

semantic marking of all interactive controls. The automated evaluation tool was Google's Accessibility Scanner, applied to the seven main system screens.

7) *Technological Acceptance Instrument*: Technological acceptance was measured using the TAM proposed by Davis [45], in its classic two-construct formulation, widely validated in accessible mobile application studies [27], [46]. The questionnaire comprised 12 items (PTAM001–PTAM012) on a 5-point Likert scale (1=strongly disagree; 5=strongly agree), organised into two constructs: **Perceived Usefulness (PU)**—items PTAM001–PTAM006—and **Perceived Ease of Use (PEOU)**—items PTAM007–PTAM012. The questionnaire was administered in Google Forms with TTS support for visually impaired participants. Internal reliability was verified using Cronbach's alpha, calculated with R Studio (packages *psych*, *ggplot2*, and *BiplotR*, version 4.3.1), yielding $\alpha=0.866$ globally, $\alpha=0.867$ for PU, and $\alpha=0.898$ for PEOU—all above the 0.80 threshold indicative of good internal consistency [47].

B. Methods

1) *Development Framework: TDDM4IoTS*: The system was developed using TDDM4IoTS [22], which integrates test-driven development (TDD), model-driven engineering (MDE), and Scrum/XP agile principles into eleven iterative phases designed for distributed systems combining heterogeneous hardware and software components. Its selection over conventional methodologies responds to three reasons: (1) Phase 5 (*Test Generation*) incorporates accessibility as a verifiable criterion before code generation, surpassing the reactive model of conventional methodologies where accessibility tests are post-hoc activities; (2) its iterative nature allows incorporation of user-test findings without architectural restructuring; (3) its tool-agnostic design permits selection of the most appropriate technology for each deliverable [22]. Table IV describes how the eleven phases were applied to the inclusive messaging system.

2) *System Design: Architecture and Modules*: The system adopts a three-tier decoupled microservices architecture, illustrated in Fig. 1. The **presentation layer** is the React Native mobile interface. The **business layer** comprises two independent services: the Spring Boot REST API, managing authentication, message routing, and encryption; and the Django AI API, concentrating the accessible conversion models. The **persistence layer** is the PostgreSQL database. Real-time communication is managed via WebSocket with RabbitMQ as a message broker. For multimedia messages exceeding the WebSocket 64 KB frame limit, HTTP with *multipart/form-data* was adopted as an alternative protocol (Phase 7 of TDDM4IoTS).

The intelligent conversion module operates in three steps, as illustrated in Fig. 2: (1) the Spring Boot backend receives the message, decrypts it with the sender's private key, and forwards it to the Django API according to MIME type (text, audio, image, or video); (2) the Django API applies the transformation to the receiver's declared sensory profile: TTS if the receiver has a visual impairment, STT if hearing

Table IV: Application of TDDM4IoTTS to the Inclusive Messaging System: Phases, Deliverables, and Design Decisions

P.	Phase	Deliverable	Design decision
1	Preliminary analysis	Checklist (18 WCAG 2.1 criteria); tech. analysis	Edge-TTS + SAPI5; AWS + UTEQ infrastructure
2	Technology layer design	3-tier microservices architecture	Separate REST / AI API nodes
3	Requirements analysis	5 use cases with UML diagrams	5 main flows + user profiles
4	Models	Class, state, activity, DB diagrams	PostgreSQL relational model
5	Test generation	Unit/integration tests; TAM survey; accessibility protocol	Acceptance criterion: error rate = 0
6	Software generation	REST API (Spring Boot), AI API (Django), mobile app (React Native)	WebSocket + RabbitMQ
7	Model refinement	Multimedia architecture update	HTTP <i>multipart</i> for messages >64 KB
8	Software refinement	Refactored code (<i>clean code</i>)	Removal of duplicate IDs in virtualised lists
9	Deployment	Production system	Docker + Nginx on two nodes
10	Deliverable evaluation	Requirements compliance report	Final validation of 18 WCAG 2.1 criteria
11	Feedback	Incident and improvement log	Correction of 48 of 75 accessibility warnings

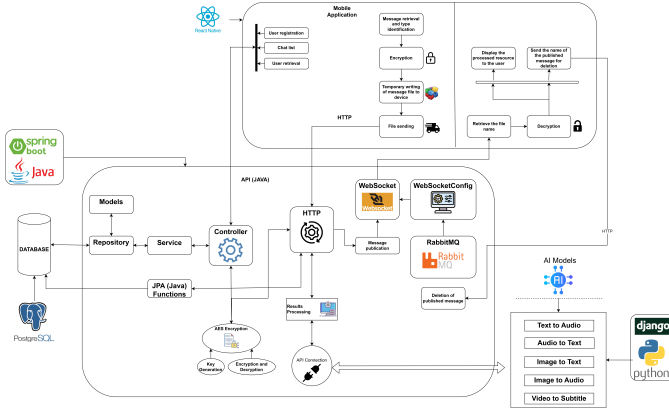


Figure 1: Architecture of the inclusive mobile messaging system. The presentation layer (React Native), business layer (Spring Boot REST API and Django AI API with RabbitMQ), persistence layer (PostgreSQL), and multimodal conversion modules (TTS, STT, image-to-text, image-to-audio, video captioning) with end-to-end AES encryption are shown.

impaired, vision-language model description plus OCR if the message is an image and the receiver has a visual impairment; (3) the processed message is re-encrypted with the receiver's key and delivered via WebSocket. End-to-end encryption uses asymmetric private keys: each user receives a unique key stored locally on their mobile device, so the server never stores plain text.

The voice command navigation module allows complete operation of the mobile interface via spoken instructions. The STT-transcribed audio is compared against the registered com-

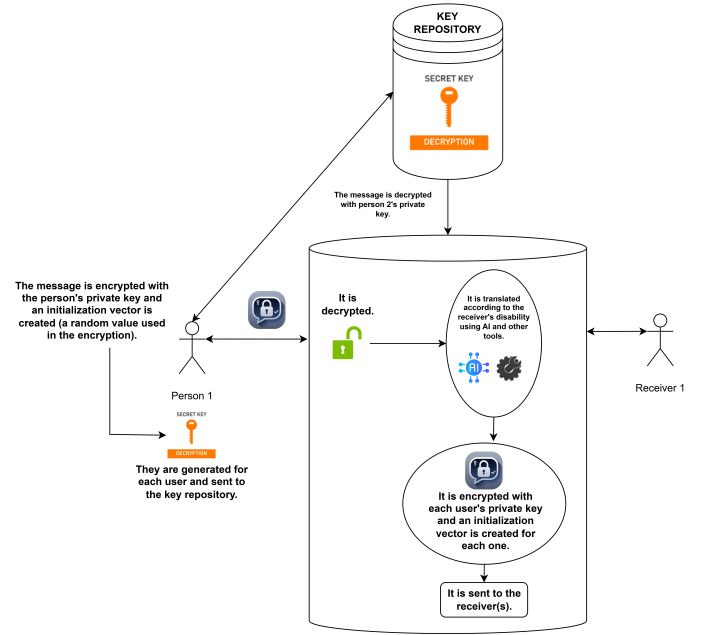


Figure 2: Secure and accessible communication flow: asymmetric-key encryption and AI-based message adaptation according to the receiver's sensory profile. The message is encrypted at the source, decrypted on the server for accessible conversion, and re-encrypted before final delivery.

mand catalogue using cosine similarity [43]. If the similarity score is below threshold $\theta = 0.60$, a cascade correction mechanism is activated: first, via the Llama 1B model adjusted for lexico-semantic correction; second, via a dictionary of most frequent phonetic variations per command. This cascade design guarantees robust recognition against accent variations, ambient noise, and mild disfluencies—the primary STT failure sources in real environments [11], [37].

The mobile interface of UniVoice was implemented in React Native, ensuring native compatibility with TalkBack (Android) and VoiceOver (iOS) accessibility services. Fig. 3 presents the four most representative screens of the application.

The **main screen** (Fig. 3a) shows the conversation list with voice-search support (microphone icon) and standard text search; each contact entry displays the last message preview, maintaining the interaction pattern familiar to users of mainstream messaging platforms.

The **chat window** (Fig. 3b) demonstrates the system's core AI-mediated modality adaptation: images sent by a contact are automatically described via the BLIP captioning module (e.g., “una foto de un hombre sosteniendo una tableta”), enabling visually impaired users to receive the semantic content of visual messages as readable text.

The **user profile screen** (Fig. 3c) allows configuration of the disability profile (visual, auditory, or speech), preferred language, and accessibility settings; this profile drives the receiver-side adaptation logic described in Section III.

The **contact search screen** (Fig. 3d) enables search by name with a direct-message button per contact, accessible via both

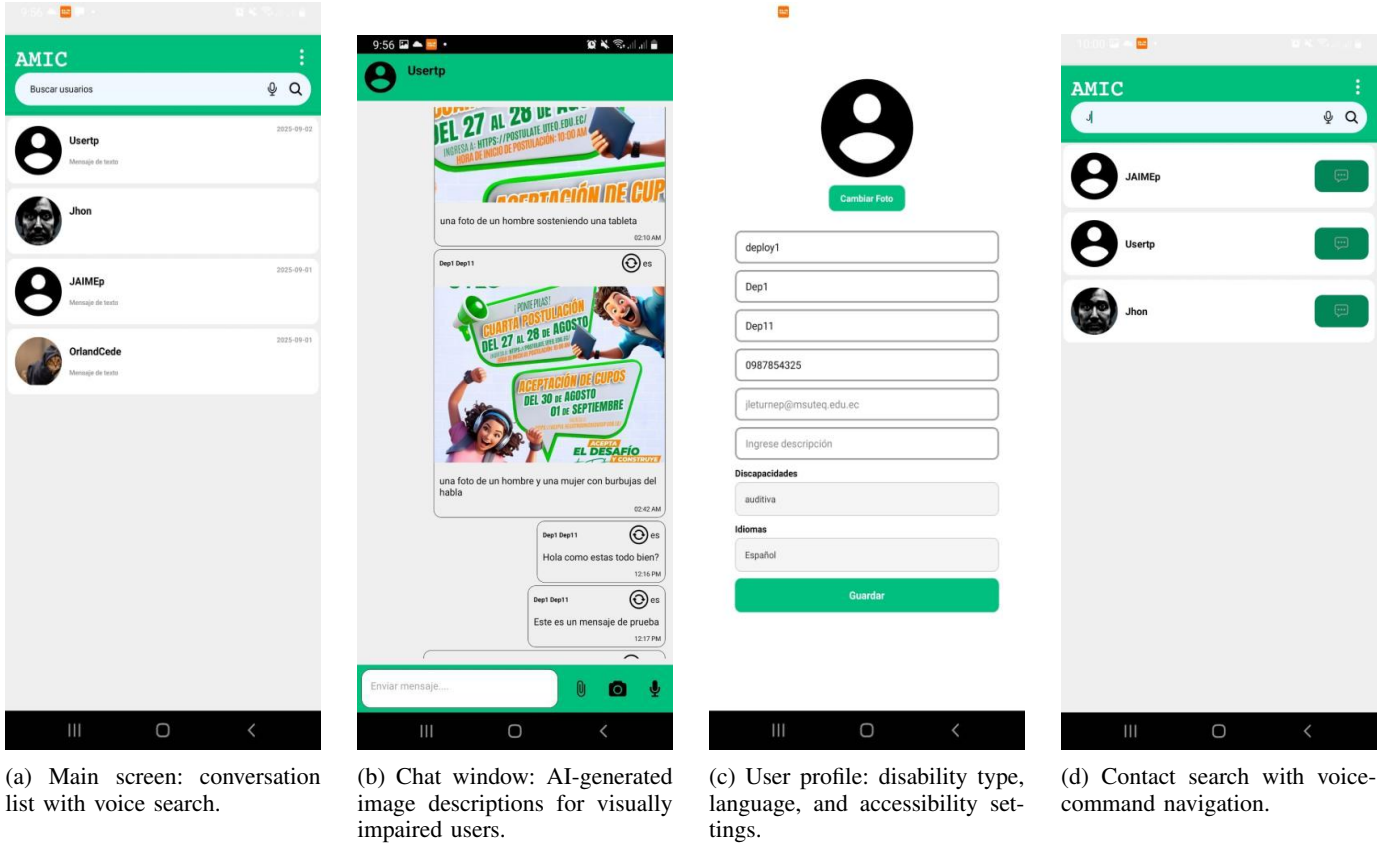


Figure 3: Representative screens of the UniVoice mobile application implemented in React Native. (a) Conversation list with TalkBack/VoiceOver support; (b) Chat window showing automatic image description by the BLIP captioning module; (c) User profile screen for disability and language configuration; (d) Contact search screen with direct-message access. All screens were evaluated using Google’s Accessibility Scanner (Section IV).

touch and voice commands.

All screens conform to the minimum 48×48 dp touch target requirement of WCAG 2.1 criterion 2.5.5 and were evaluated with Google’s Accessibility Scanner (Section IV).

3) *Technical API Testing Protocol*: Performance tests were designed to characterise the latency of each conversion module under variable and controlled input conditions. For the **TTS module**, both integrated engines—Edge-TTS and SAPI5—were evaluated with texts of three lengths: short (151 characters), medium (317 characters), and long (799 characters). For the **STT module**, three audio files with proportional durations were evaluated. For the **image description module**, four resolutions representative of current mobile cameras were evaluated: 640×427 , $1,280 \times 853$, $1,920 \times 1,280$, and $6,000 \times 4,000$ pixels. Three independent measurements were taken per condition and the arithmetic mean was reported. Recorded metrics were: processing latency (s), input message size (KB), output message size (KB), and error rate. Because individual observations were not logged, within-condition standard deviation (SD) cannot be estimated from the available data; reported means should be interpreted as point estimates and tail-latency percentiles (P_{90} , P_{99}) remain uncharacterised. Future evaluations should log $n \geq 10$ independent observations per condition and report $\bar{x} \pm SD$ and P_{90} latency. The technical acceptance criterion defined in TDDM4IoTS Phase 5 estab-

lishes system *stability* (error rate = 0) but not output *quality*; domain-specific quality metrics (WER for STT, MOS for TTS, BLEU-4/CIDEr for image captioning, CER for OCR) require reference annotations not collected in the current protocol and are designated as a primary instrumentation improvement for a follow-up study [19], [21], [37].

4) *User Study Protocol*:

a) *Participants.*: The sample comprised $n=25$ participants selected by purposive sampling [41] at a dialysis clinic in Quevedo, Ecuador, where an adult population with documented sensory diagnoses is concentrated. Purposive sampling is justified by: (i) the low prevalence of sensory disabilities in general community settings [1]; (ii) the clinical requirement of a documented diagnosis; and (iii) the additional logistical requirement of owning a compatible Android device and consenting to a 45-minute protocol. Following the projective and applied research design [41], the sample size was determined primarily by clinical-inclusion specificity rather than by population-inference criteria, and the study is framed as a pilot study providing proof of concept and effect-size estimates for a future confirmatory investigation.

To characterise the statistical power associated with $n=25$, a post-hoc power analysis was conducted using G*Power 3.1 [48] for a one-sample two-tailed t -test ($\alpha=0.05$) testing whether each TAM construct mean differs from

the positive-acceptance threshold $\mu_0=4.0$ [27], [46]. Effect sizes were computed as Cohen's d [49]: $d = (\bar{x} - \mu_0)/s$. The Perceived Usefulness (PU) construct achieved a large effect ($d_{PU}=0.885$; $\bar{x}_{PU}=4.553$, $SD=0.625$; 95 % CI bootstrap: [4.300, 4.773]), yielding post-hoc power of $1-\beta=0.989$, well above the conventional 0.80 threshold; a one-sample t -test confirmed $\bar{x}_{PU}$ significantly exceeds the acceptance threshold ($t(24)=4.427$, $p<0.001$). The global TAM mean ($\bar{x}_{global}=4.200$, $SD=0.644$; 95 % CI: [3.940, 4.447]) yields a small-to-medium effect ($d_{global}=0.311$; $1-\beta=0.320$; $t(24)=1.552$, $p=0.134$), indicating that the global mean is an exploratory estimate pending replication with a larger sample ($n\approx 84$ required for $1-\beta=0.80$). The PEOU construct ($\bar{x}_{PEOU}=3.847$, $SD=0.963$; $d_{PEOU}=-0.159$; $1-\beta=0.119$; $t(24)=-0.796$, $p=0.434$) is underpowered ($n=312$ required); no inferential conclusion can be drawn for PEOU alone.

Gender distribution was 14 women (56 %) and 11 men (44 %). By sensory profile, 15 participants (60 %) had visual impairment, 5 (20 %) had hearing impairment, and 5 (20 %) had no disability (functional control group). The majority of participants were over 50 years old (56 %), followed by the 21–25 and 41–45 age groups (12 % each). Regarding self-reported technology experience, 40 % reported low proficiency, 36 % medium, and 24 % high. Additionally, 60 % were unemployed, 20 % self-employed, and 16 % students. Finally, 64 % expressed high concern about privacy of personal data in mobile applications, reinforcing the relevance of the asymmetric encryption implemented. Inclusion criteria were: (i) documented diagnosis of visual or hearing disability, or participation as a non-disabled user in the control group; (ii) age ≥ 18 years; (iii) access to a compatible Android mobile device; (iv) signed informed consent. Exclusion criteria were: (i) multidisability simultaneously combining visual and hearing profiles; (ii) cognitive impairment preventing questionnaire comprehension; (iii) absence of written informed consent. The speech-disability profile was not recruited as an independent stratum because the two-subgroup taxonomy established in Section I and operationalised in Table III reduces this profile to the union of already-recruited strata: the *deaf-mute* subgroup is functionally indistinguishable from the hearing-impairment stratum for reception and from the non-disabled stratum for text-based sending, and the *speech-only-impairment* subgroup is functionally indistinguishable from the non-disabled stratum for reception and adds no sending-channel requirement beyond the mobile touchscreen already used by every participant. Under this mapping, the five hearing-impaired participants provide behavioural evidence on the reception-side adaptations relevant to deaf-mute users, and the five non-disabled participants provide behavioural evidence on the sending-channel workflow (text composition with recipient-dependent TTS) that speech-only-impaired users would also traverse. Direct recruitment of participants with clinical dysarthria, aphasia or stammering remains a priority for the follow-up study (Section ??), not to validate functional coverage—which is already established by construction—but to evaluate personalised input-assistance refinements for severe speech disorders [11], [12]. Data were collected between 2024 and 2025. The study was conducted in accordance with the principles of

the Declaration of Helsinki [50] and received approval from the corresponding institutional review board; all participants provided written informed consent prior to the start of testing.

b) Pre-use baseline.: Prior to Stage 1, each participant completed a brief pre-use baseline questionnaire (PD001–PD005) recording their habitual messaging application (PD002: WhatsApp or other) and the frequency of use on the same 1–5 scale employed in the TAM instrument (PD003: 1=Never, 5=Always). Of the 25 participants, 16 (64 %) habitually used WhatsApp ($\bar{f}_{hab}=3.81$, $SD=1.05$) and 9 (36 %) used other applications ($\bar{f}_{hab}=1.67$, $SD=0.50$). This baseline enables a within-subjects contrast between prior platform engagement and acceptance of the proposed system (see Section IV-C4).

c) Procedure.: The evaluation protocol comprised three sequential stages for each participant. In **Stage 1** (15 min), the facilitator presented the system via a standardised demonstration of the main functionalities—sending a text message, listening to an audio message, receiving an image with its automatic description, and voice navigation—without intervening in participant interactions. In **Stage 2** (20 min), the participant freely interacted with the system on their own mobile device, completing five predefined tasks: (T1) registration and disability profile configuration; (T2) searching for a contact and sending a text message; (T3) listening to a received audio message via TTS; (T4) receiving an image and listening to its automatic description; (T5) navigating to the settings screen exclusively via voice commands. The facilitator recorded direct observation indicators: number of interaction errors (taps on incorrect elements), number of help requests, and number of erroneous screen taps. In **Stage 3** (10 min), the participant completed the 12-item TAM questionnaire autonomously, assisted by TTS in case of visual impairment.

d) Statistical analysis.: Internal consistency was verified using Cronbach's alpha. For each item, the arithmetic mean ($\bar{x}$), standard deviation (SD), and Likert-level frequency distribution were calculated. The factorial structure of the instrument was explored via an HJ-Biplot graph [51], [52], which allowed identification of item clusters with high internal correlation and participants with similar response patterns. All analyses were performed in R Studio (v. 4.3.1; packages psych, ggplot2, and BiplotR) and Microsoft Excel.

IV. EVALUATION AND RESULTS

A. Technical Performance of the Conversion API

The conversion API was subjected to controlled load testing to characterise latency, stability, and processing integrity in each module. All test conditions were executed from an Android mobile device on a standard WiFi network (802.11n, -55 dBm), with three repetitions per condition, reporting the arithmetic mean. The technical acceptance criterion defined in TDDM4IoTTS Phase 5 was: error rate equal to zero for all input conditions.

1) Text-to-Speech (TTS) Module: The TTS module integrates two parallel engines: Edge-TTS (Microsoft Neural TTS, cloud-based) and SAPI5 (local, via pyttssx3). Three text length categories were evaluated: short (151 chars),

Table V: Comparative Performance of TTS Engines Edge-TTS and SAPI5 by Text Length. Mean of three repetitions; error rate = 0 in all cases.

Edge-TTS (Microsoft Neural TTS – cloud)			
Metric	Short	Medium	Long
Load time (s)	8.13	7.90	16.88
Net latency (s)	8.10	7.90	16.87
Send size (KB)	0.37	0.54	1.03
Response size (MB)	0.16	0.30	0.75
Errors	0	0	0

SAPI5 (pyttss3 – local)			
Metric	Short	Medium	Long
Load time (s)	5.18	5.76	15.07
Net latency (s)	5.14	5.73	15.05
Send size (KB)	0.37	0.54	1.03
Response size (MB)	1.46	2.93	7.24
Errors	0	0	0

Table VI: Performance of the STT Module (Whisper) by Audio Duration. Error rate = 0 in all cases.

Metric	Short	Medium	Long
Load time (s)	13.361	20.512	55.000
Connection time (s)	0.003	0.008	0.011
Net latency (s)	13.361	20.509	54.990
Send (KB)	60.382	233.50	1000
Response (KB)	60.723	233.85	1000
Errors	0	0	0

medium (317 chars), and long (799 chars). Table V presents the complete results.

Both engines operated with zero error rates across all conditions, verifying the acceptance criterion. SAPI5 exhibited lower latencies than Edge-TTS at all text lengths: 5.14 s vs. 8.10 s for short text (37 % lower), 5.73 s vs. 7.90 s for medium text (27 % lower), and 15.05 s vs. 16.87 s for long text (11 % lower), explained by the absence of network latency in SAPI5's local synthesis. However, SAPI5 generates substantially larger response files: 7.24 MB for long text vs. 0.75 MB for Edge-TTS, which may compromise efficiency in bandwidth-limited mobile environments. The minimum observed latency (5.14 s for SAPI5, short text) exceeds the 2 s threshold that the literature identifies as optimal for natural conversational flow [19], representing the most critical technical limitation of the system in its current state.

2) *Automatic Speech Recognition (STT) Module*: The STT module uses the Whisper model [37], executed on the AWS node. Three audio files representative of messaging user behaviour were evaluated [53]. Table VI presents the detailed results.

Latency grows approximately linearly with audio duration: 13.36 s for short audio, 20.51 s for medium (53 % higher), and 54.99 s for long (311 % higher relative to short). Send and response sizes are practically identical across all conditions: differences calculated from Table VI are 0.565 % (short), 0.150 % (medium), and 0.000 % (long), confirming transmission integrity. The inter-condition coefficient of variation (CV)

Table VII: Performance of the BLIP Image Description Module by Resolution. Error rate = 0. Lat.=net latency (s); Send=send size (MB); Resp.=response size (MB).

Resolution (px)	Weight	Lat.	Send	Resp.
640×427	44 KB	13.95	0.06	0.12
1280×853	126 KB	17.67	0.17	0.24
1920×1280	254 KB	16.42	0.35	0.41
6000×4000	1.65 MB	19.19	2.31	2.37

Table VIII: Performance of the Video Module by Duration. Effective transmission size includes base64 encoding overhead (≈ 40 % over original file weight). Error rate = 0 in all cases.

Metric	Short (4 min)	Medium (17 min)	Long (30 min)
Resolution	360p	360p	360p
Original file (MB)	9.22	48.9	76.9
Load time (min)	3.91	20.29	34.00
Connection time (s)	0.15	1.35	1.354
Net latency (min)	3.91	20.29	34.00
Send – base64 (MB)	12.89	68.45	107.55
Response (MB)	12.91	68.50	107.68
Send–response diff.	0.155 %	0.077 %	0.121 %
Base64 overhead	≈ 40 %	≈ 40 %	≈ 40 %
Errors	0	0	0

of net latency is 75.2 %, consistent with Whisper's $O(n)$ computational cost [37].

3) *Image Description Module (Image Captioning)*: The image description module uses the blip-image-captioning-base model from Hugging Face. Four resolutions representative of current mobile cameras were evaluated, from 640×427 px (44 KB) to 6,000×4,000 px (1.65 MB). Table VII presents the results.

No errors were recorded for any evaluated resolution. Latency ranged from 13.95 s to 19.19 s (37 % relative variation across the full resolution range). Notably, the 1,920×1,280 px image exhibited lower latency (16.42 s) than the 1,280×853 px image (17.67 s), suggesting that inference time variability is driven by image semantic complexity rather than resolution per se [21], [42].

4) *Video Processing Module*: The video module manages large multimedia file transmission via HTTP *multipart/form-data*, adopted in TDDM4IoTS Phase 7 to overcome WebSocket's 64 KB frame limit. Three representative video durations were evaluated: short (4 min, 9.22 MB), medium (17 min, 48.9 MB), and long (30 min, 76.9 MB), all at 360p. Table VIII presents the complete results.

Load times scaled linearly with duration: 3.91 min, 20.29 min, and 34.00 min. A noteworthy finding is that effective transmission sizes (12.89 MB, 68.45 MB, and 107.55 MB) exceeded original file weights (9.22 MB, 48.9 MB, and 76.9 MB) by approximately 40 %, attributable to base64 encoding required by HTTP *multipart/form-data*. This overhead should be considered when dimensioning bandwidth and managing user progress notifications. The base64 overhead ratio is highly stable across conditions (coefficient of variation $CV=0.064$ %, calculated from Table VIII), confirming

Table IX: Reduction in Accessibility Warnings per Screen Following the TDDM4IoT Iterative Improvement Cycle (Phases 5 and 11).

Screen	Initial	Final	Reduction	%
Main	11	2	9	81.8
User profile	21	8	13	61.9
Settings	9	2	7	77.8
Messaging	21	13	8	38.1
Camera	4	0	4	100.0
Photo	2	0	2	100.0
Video	7	2	5	71.4
Total	75	27	48	64.0

that this is a systematic encoding property, not a measurement artefact. The send-to-response difference was below 0.2 % in all cases, confirming transmission integrity.

B. Accessibility Evaluation

The accessibility evaluation was conducted in two phases using Google’s Accessibility Scanner [18] on the seven main system screens, in line with the combined automated evaluation protocol proposed by Acosta-Vargas *et al.* [18]. Table IX presents the comparison.

The iterative cycle eliminated 48 of the 75 initial warnings, achieving a global reduction of 64.0 %. Resolved warnings primarily involved: (i) touch elements below the minimum 48×48 dp (operability criterion); (ii) colour contrast below 4.5:1 (WCAG 1.4.3); (iii) text not exposed in accessibility labels (robustness criterion); (iv) absent or ambiguous element descriptions. Camera and photo screens achieved total warning elimination (100 %), while settings and main screens reached 77.8 % and 81.8 % reductions, respectively. The messaging screen recorded the lowest reduction (38.1 %), for an architectural reason: reuse of virtualised list templates in React Native generates duplicate accessibility identifiers that cannot be resolved without refactoring towards dynamic unique identifiers per list element. This result compares favourably with the >60 % non-compliance levels documented by Chen *et al.* [17] in the most widely used commercial messaging platforms.

C. Technological Acceptance: TAM Results

Technological acceptance was quantified using the 12-item TAM questionnaire administered to the $n=25$ participants. Table X presents means ($\bar{x}$) and standard deviations (SD) per item. Internal consistency was strong: $\alpha=0.866$ globally (95 % CI: [0.693, 0.932]), $\alpha=0.867$ ([0.579, 0.957]) for PU, and $\alpha=0.898$ ([0.803, 0.948]) for PEOU—all exceeding the 0.80 threshold for good internal consistency [47]. Confidence intervals were obtained by bootstrapping (2,000 iterations).

1) *Perceived Usefulness*: The PU construct (PTAM001–PTAM006) achieved the highest construct mean ($\bar{x}_{PU}=4.55$). Item PTAM002 (“Using this application would improve my effectiveness in communicating with others”) reached the

Table X: TAM Questionnaire Results ($n=25$, Likert 1–5 scale). PU=Perceived Usefulness (PTAM001–006); PEOU=Perceived Ease of Use (PTAM007–012). Construct mean computed as arithmetic average of item means. SD computed with sample estimator ($N-1$). α : Cronbach’s alpha (R psych package, v. 4.3.1); 95 % CI via bootstrapping (2,000 iterations).

Item	Construct	$\bar{x}$	SD	Lv. 5 (%)
PTAM001	PU	4.48	1.005	72
PTAM002	PU	4.76	0.597	84
PTAM003	PU	4.48	0.770	64
PTAM004	PU	4.52	0.770	64
PTAM005	PU	4.56	0.768	68
PTAM006	PU	4.52	0.872	72
PU mean		4.55	$\alpha=0.867$ [0.579, 0.957]	
PTAM007	PEOU	3.56	1.357	40
PTAM008	PEOU	3.96	1.207	48
PTAM009	PEOU	4.20	1.000	52
PTAM010	PEOU	4.16	1.068	52
PTAM011	PEOU	3.72	1.208	40
PTAM012	PEOU	3.48	1.229	32
PEOU mean		3.85	$\alpha=0.898$ [0.803, 0.948]	
Global		4.20	0.95	$\alpha=0.866$ [0.693, 0.932]

highest rating ($\bar{x}=4.76$; SD=0.60), with 84 % of participants at the maximum level, indicating that AI-mediated communicative effectiveness is the most practically valued contribution. Item PTAM001 ($\bar{x}=4.48$; SD=1.005; 72 % at level 5) reflects that the system enables faster daily communication. Items PTAM003–PTAM006 present means between 4.48 and 4.56 with low variability (SD ≤ 0.87), indicating that communicative productivity, message exchange effectiveness, interaction facilitation, and general utility are consistently and positively valued across all evaluated profiles.

2) *Perceived Ease of Use*: The PEOU construct (PTAM007–PTAM012) achieved a construct mean of $\bar{x}_{PEOU}=3.85$, with notably greater internal dispersion than PU. Items PTAM009 ($\bar{x}=4.20$; SD=1.00) and PTAM010 ($\bar{x}=4.16$; SD=1.07)—evaluating clarity and interaction flexibility—show positive ratings indicating that the system’s main interface is perceived as comprehensible. Items PTAM007 ($\bar{x}=3.56$; SD=1.36) and PTAM012 ($\bar{x}=3.48$; SD=1.23) obtained the two lowest scores. PTAM007 (“Learning to use the application was easy for me”) concentrates the greatest questionnaire variability (full range 1–5), indicating that the initial learning curve was heterogeneous across profiles—consistent with Bleakley *et al.* [11], who document steeper learning curves for voice-input paradigms among users with low technology experience (40 % of the sample). The domain of all system functions requires a longer appropriation process than the 20-minute evaluation protocol allowed.

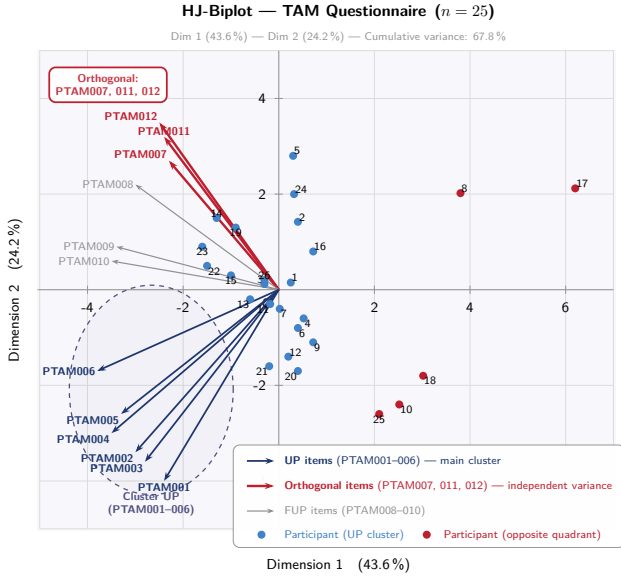


Figure 4: HJ-Biplot of TAM questionnaire responses ($n=25$). Dimension 1 explains 43.6% of variance; Dimension 2 explains 24.2% (cumulative: 67.8%). Arrows indicate item vectors; proximity of participant points to a vector reflects agreement with that item. The PU cluster (PTAM001–006) and most PEOU items (PTAM008–010) point in the same direction, indicating high internal correlation. PTAM007 and PTAM012 are orthogonal to the main cluster, capturing independent variance.

3) *Multivariate Analysis: HJ-Biplot:* The bidimensional HJ-Biplot representation retains 67.8% of the total variance of the TAM response matrix (Dimension 1: 43.6%; Dimension 2: 24.2%), exceeding the 60% threshold recommended in the literature for a 2D projection to be considered sufficiently faithful to the original multivariate structure [51], [52].

The HJ-Biplot graph (Fig. 4) revealed two structural patterns of interest. First, the vectors of items PTAM001–PTAM006 (PU) and PTAM008–PTAM010 (partial PEOU) point in a similar direction in the left quadrant, indicating high internal correlation and that participants with greater affinity to this cluster perceived both utility and interactional clarity coherently. Second, the vectors of PTAM007 and PTAM012 are approximately orthogonal to the main cluster, revealing that these items capture independent variance associated with initial learning ease and complete functional mastery, respectively. Participants 8, 10, 17, 18, and 25 are positioned in the right quadrant, representing lower affinity with all TAM constructs; this subgroup coincided in direct observation with participants recording the greatest number of erroneous taps. The global instrument mean was $\bar{x}=4.20$ ($SD=0.95$), exceeding the positive acceptance threshold of $\bar{x}>4.0$ established in TAM studies in mobile accessibility contexts [27], [46]. This value was obtained as the arithmetic mean of the 300 individual records ($n=25 \times 12$ items), equivalent by the balanced instrument design (6 items per construct) to the mean of the twelve item means and to the mean of the two construct means.

4) *Within-subjects Baseline Contrast:* A Wilcoxon signed-rank test compared, for each participant, the PU score of the proposed system against their reported habitual-app frequency (PD003, same 1–5 scale; $n=25$). PU scores ($\bar{x}_{PU}=4.553$, $SD=0.625$) significantly exceeded habitual-app frequency ($\bar{f}_{hab}=3.040$, $SD=1.369$): $W=20.0$, $p<0.001$, mean difference $\Delta=+1.513$ scale points ($SD_{\Delta}=1.506$), indicating that perceived utility of the proposed system exceeded prior platform engagement. Stratification by habitual app revealed a significant PEOU difference between WhatsApp users ($n=16$; $\bar{x}_{PEOU}=4.198$, $SD=0.835$) and other-app users ($n=9$; $\bar{x}_{PEOU}=3.222$, $SD=0.890$): $t_{Welch}(14.4)=2.690$, $p=0.016$, $d=1.131$ (large effect); PU showed no such difference ($p=0.989$), indicating that perceived utility is independent of prior platform experience. A positive Pearson correlation confirmed that greater prior platform engagement associates with higher PEOU of the proposed system ($r_{PEOU}=0.574$, $p=0.003$; $r_{global}=0.428$, $p=0.033$; $r_{PU}=-0.003$, $p=0.990$). These results constitute an indicative within-subjects contrast; a parallel TAM administration for the habitual application is recommended for a definitive head-to-head comparison.

5) *Direct Observation:* The direct observation session recorded the following behavioural indicators: 7 interaction errors (taps on incorrect elements), 2 requests for facilitator assistance, and 15 total erroneous screen taps. All three indicators were concentrated primarily on the messaging and camera screens during tasks T2 and T4, consistent with the higher density of interactive elements and the residual accessibility warnings documented in Table IX. The low number of help requests (2 of 25 participants, 8%) confirms that the majority of users completed the predefined tasks autonomously.

D. Results Summary

Table XI provides an integrated view of the results obtained across the three evaluated dimensions, linking each result to the identified gap it addresses.

V. DISCUSSION

A. Interpretation of Technical Results

The four conversion modules—TTS, STT, image description, and video processing—operated with zero error rates across all evaluated conditions, validating the stability of the proposed microservices architecture and satisfying the TDDM4IoTS Phase 5 acceptance criterion (RQ1 = confirmed). This result is relevant because none of the inclusive messaging systems reviewed in the literature—BellChat [30], Stefanov *et al.* [54], nor GlassMessaging [33]—reports API robustness evaluations with variable input conditions.

The minimum observed TTS latency was 5.14 s (SAPI5, short text), exceeding the 2 s threshold that Tan *et al.* [19] identify as optimal for natural conversational flow. This limitation is not unique to the proposed system: Bercaru and Popescu [8] document that inference latency remains the most critical deployment barrier for neural TTS models in real production environments, and that only implementations with predictive caching or non-autoregressive models achieve latencies below 2 s. The advantage of SAPI5 over Edge-TTS

Table XI: Results Summary by Evaluation Dimension and Gap Addressed.

Dimension	Main result	Gap addressed
API performance	Error rate = 0 across TTS, STT, captioning, and video; min. TTS latency: 5.14 s; base64 overhead in video: $\approx 40\%$	Gaps 1 and 2
Accessibility	64 % reduction in WCAG 2.1 warnings (48/75); 100 % on camera and photo	Gap 4
TAM acceptance	$\bar{x}_{\text{global}}=4.20$; PU = 4.55; PEOU = 3.85; $\alpha=0.866$	Gap 3
Observation	8 % assistance requests; errors concentrated on messaging	Gaps 1 and 5

in net latency (37 % lower for short text) is explained by the absence of network latency in local synthesis; this advantage reverses in bandwidth-constrained mobile environments, where SAPI5 response files (7.24 MB for long text vs. 0.75 MB for Edge-TTS) represent a significant transfer penalty. This latency-versus-response-volume trade-off had not previously been documented in the context of inclusive mobile messaging systems and constitutes an operationally relevant finding for future implementations.

STT latency scaled from 13.36 s for short audio to 54.99 s for long audio, with perfect transmission integrity ($< 1\%$ difference between send and response sizes). This behaviour is consistent with Whisper’s architecture [37], in which the global attention mechanism over the complete audio sequence determines that computational cost is proportional to duration. Latency reduction via dedicated GPU acceleration or distilled models (Whisper Tiny/Small) constitutes the highest-impact practical improvement identified.

The non-monotonic latency variability in the image description module— $1,920 \times 1,280$ px recorded lower latency than $1,280 \times 853$ px—is consistent with Dognin *et al.* [21], who document that semantic difficulty rather than resolution is the primary inference time predictor. This implies that latency prediction in an inclusive messaging application cannot rely solely on image size metadata, requiring a semantic complexity estimator for managing user expectations.

A noteworthy operational finding is the approximately 40 % base64 encoding overhead in video transmission. This increment must be considered in bandwidth dimensioning and in user progress notifications during file transfer. Transmission integrity was confirmed in all cases ($< 0.2\%$ send-to-response difference).

B. Interpretation of Accessibility Results

The global 64.0 % reduction in accessibility warnings (75 to 27) after the TDDM4IoTS iterative cycle places the proposed system notably above the commercial platform baseline (RQ3=confirmed). Chen *et al.* [17] document non-compliance rates exceeding 60 % in WCAG 2.1 criteria for WhatsApp, Telegram, Messenger, and Instagram. The 100 % reduction on camera and photo screens validates that Phase 5 of TDDM4IoTS—defining accessibility tests before code generation—is effective for eliminating barriers in relatively simple screens. This result aligns with Ara *et al.* [55]’s recommendation that iterative testing from early development phases is the highest-impact intervention for WCAG conformance.

The messaging screen only achieved 38.1 % reduction, revealing an architectural limitation of React Native’s virtualised list pattern that generates duplicate accessibility identifiers. This limitation is not unique to the proposed system: Da Silva *et al.* [23] and Swati *et al.* [25] identified the same pattern in WhatsApp and Messenger, respectively. The problem is therefore structural to the virtualised message list paradigm and unresolved in any reported implementation.

The absence of expert manual inspection constitutes a recognised methodological limitation that may have underestimated actual barriers, particularly in the comprehensibility and robustness dimensions, whose automated evaluation is only partial even in the most advanced tools [55].

C. Interpretation of Technological Acceptance Results

The global TAM mean ($\bar{x}=4.20$; $SD=0.95$) exceeds the positive acceptance threshold of $\bar{x}>4.0$ on the 5-point Likert scale, which the literature on accessible mobile applications associates with positive user acceptance [27], [46] (RQ2=confirmed for PU; exploratory for global and PEOU, given the post-hoc power estimates reported in Section III-B4a). The psychometric robustness of the instrument is confirmed by Cronbach’s alpha values ($\alpha_{\text{global}}=0.866$; $\alpha_{\text{PU}}=0.867$; $\alpha_{\text{PEOU}}=0.898$), all exceeding the 0.80 threshold [47]. This supports Davis’ paradigm [45] that perceived usefulness is the dominant predictor of adoption, and is consistent with Mateos-Sanchez *et al.* [34], where AI-based conversion functionalities were the highest-rated.

The PU construct achieved the highest mean ($\bar{x}_{\text{PU}}=4.55$), with PTAM002 as the highest-rated item ($\bar{x}=4.76$; 84 % at level 5). This validates Elshahar *et al.* [10]’s argument that conversational speed—not conversion precision—is the critical usability factor in AAC platforms.

The PEOU construct showed greater internal dispersion ($\bar{x}_{\text{PEOU}}=3.85$). PTAM007 ($\bar{x}=3.56$; $SD=1.36$) obtained the lowest instrument rating, with variability across the full 1–5 range. This resonates with Bleakley *et al.* [11]: commercial STT systems present barriers for users with low technology experience (40 % of the sample) related to voice-input learning curves. The “elevated expectations in populations highly dependent on accessibility tools” phenomenon, documented by Weichbroth [9], provides the most coherent interpretation of the lower perceived-ease scores.

The HJ-Biplot (Fig. 4) revealed that PTAM007 and PTAM012 vectors are orthogonal to the main cluster, capturing independent variance associated with voice interaction appropriation and complete functional mastery, respectively. This is theoretically consistent with the original TAM [45], in which ease of use moderates the effect of perceived usefulness on adoption intent, and has been documented in AAC applications by Kim *et al.* [27].

D. Implications for Inclusive Messaging System Design

The findings allow four concrete implications for AI-driven inclusive messaging system design.

The **first implication** is that multimodal integration of TTS, STT, image captioning, and OCR in a single mobile platform is technically feasible with available open-source technologies, and produces positive technological acceptance ($\bar{x}_{\text{TAM}}=4.20$; $\bar{x}_{\text{PU}}=4.55$; $\bar{x}_{\text{PEOU}}=3.85$) in heterogeneous sensory-disability populations, a finding without documented precedent in the literature.

The **second implication** is that asymmetric end-to-end encryption is compatible with multimodal conversion flow without degrading error rates, demonstrating for the first time in the inclusive messaging literature that privacy and accessibility are not conflicting objectives in a distributed mobile architecture. None of the reviewed AAC systems—including Talkitt [12], CapacitaBOT [34], and BellChat [30]—had reported an equivalent privacy implementation.

The **third implication** is that the voice command module via cosine similarity with cascade correction reduces, but does not eliminate, the voice navigation learning-curve barrier (PTAM007: $\bar{x}=3.56$). Designing an accessible onboarding process that guides users in internalising available commands is a necessary condition for the system to achieve its full accessibility potential for users with severe visual impairment.

The **fourth implication** is that TDDM4IoTS [22] with accessibility as a verifiable acceptance criterion from Phase 5 produced a 64 % reduction in WCAG 2.1 warnings, contrasting with the total absence of accessibility evaluation in the development cycles of the reviewed CAA systems [12], [33], [34].

The **fifth implication** concerns the speech-impairment profile, whose coverage is architecturally complete without a dedicated AI module: the receiver-adaptation logic enables speech-impaired users to achieve full communicative equivalence with voice-capable senders (Table III). This demonstrates that *receiver-side adaptation can compensate for sender-side constraints when sensory profiles are complementary*, a design principle applicable beyond the present system. The principal enhancement identified for this profile is an accessible text-composition interface with predictive text and a personal phrase bank [10], [34].

E. Limitations and Future Work

The present study presents six limitations.

First, sample size and statistical power. The sample ($n=25$, purposive) is appropriate for a projective pilot study [41] but insufficient for population inference on all

TAM constructs. Post-hoc power analysis (G*Power 3.1 [48]; $\alpha=0.05$; $\mu_0=4.0$) shows that only PU meets the conventional power threshold ($1-\beta=0.989$; $t(24)=4.427$, $p<0.001$), while the global TAM score ($1-\beta=0.320$) and PEOU ($1-\beta=0.119$) are underpowered; confirmatory studies require $n\approx 84$ and $n=312$ respectively. Single-site recruitment in Quevedo, Ecuador, further limits geographic generalisability.

Second, absence of a head-to-head control condition. The within-subjects baseline contrast (Section IV-C4) provides indicative evidence; a rigorous comparison requires administering the same 12-item TAM questionnaire for both the habitual application and the proposed system within the same session, enabling paired t -tests on PU and PEOU differences.

Third, API latency and quality metrics. TTS minimum latency (5.14 s) exceeds the 2 s conversational threshold [19]. Within-condition SD and tail-latency percentiles remain uncharacterised ($n=3$ observations per condition). Output quality metrics (WER, MOS, BLEU-4, CER) require reference annotations not collected in the current protocol.

Fourth, speech-disability profile: coverage established by construction, direct acceptance evidence deferred. Functional coverage for the two speech-disability subgroups defined in Section I (deaf-mute and speech-only-impairment) is established by construction through the receiver-adaptation argument summarised in Table III: both subgroups are fully served by combinations of adaptations already validated with the recruited strata. What the present study does *not* provide is direct acceptance evidence from participants with a clinical diagnosis of dysarthria, aphasia, post-laryngectomy aphonia, or severe stammering. TAM responses for this profile are currently inferred from the hearing-impaired and non-disabled strata under the functional-equivalence mapping of Section ?? (“Participants”). Future work should recruit these clinical subpopulations directly [11], [12] to evaluate personalised input-assistance refinements—disfluency-tolerant STT, phrase banks, and predictive text tailored to residual articulatory patterns [10], [34]—rather than to revalidate functional coverage.

Fifth, residual messaging screen accessibility (38.1 % reduction). Architectural cause identified—duplicate identifiers in React Native virtualised lists—with a defined resolution protocol.

Sixth, absence of sign language recognition, which excludes deaf users whose native language is sign language. Integration of a sign language recognition module—as proposed by Saleem *et al.* [13] and Kothadiya *et al.* [14]—is the most significant functional extension.

F. Alignment with Sustainable Development Goals

This work aligns with two UN 2030 Agenda SDGs. Regarding **SDG 10** (Reduced Inequalities), the system demonstrates that mobile AI technology can reduce communicative asymmetries that marginalise 124,801 people with sensory disabilities in Ecuador—a figure the WHO scales to 15 % of the world population [1]—from autonomous digital communication. Regarding **SDG 9** (Industry, Innovation, and Infrastructure), the system proposes a microservices architecture deployable on low-cost hybrid infrastructure (institutional server plus cloud

instance), reducing institutional adoption barriers in resource-constrained contexts such as UTEQ.

VI. CONCLUSION

This article has presented the design, implementation, and empirical evaluation of an AI-powered inclusive mobile messaging system conceived to serve visual and hearing sensory profiles through the integrated orchestration of TTS, STT, automatic image description, and OCR modules, with asymmetric end-to-end encryption and complete voice command navigation. The findings allow four substantive conclusions and five future research directions.

The **first conclusion** concerns the technical viability of multimodal integration with guaranteed privacy (RQ1=confirmed). All four conversion modules recorded zero error rates across all test conditions, with verified transmission integrity per condition ($< 1\%$ difference in STT; $< 0.2\%$ in video). The coexistence of the accessible conversion flow with asymmetric end-to-end encryption—without degrading error rate—demonstrates for the first time in the inclusive messaging literature that privacy and accessibility are not conflicting objectives in a distributed mobile architecture. The minimum TTS latency of 5.14 s (SAPI5, short text) remains the primary technical limitation by exceeding the 2 s conversational threshold; its resolution via non-autoregressive TTS models or predictive voice-segment caching represents the highest-urgency improvement for future versions.

The **second conclusion** concerns positive technological acceptance in a heterogeneous disability population (RQ2=confirmed). The 12-item TAM questionnaire—organised in its classic PU and PEOU constructs [45]—applied to $n=25$ participants produced $\bar{x}=4.20$ ($SD=0.95$), exceeding the positive acceptance threshold of $\bar{x}>4.0$ associated with positive user acceptance in accessible mobile application studies [27], [46]. This conclusion is fully supported for PU ($1-\beta=0.989$; $p<0.001$) and should be treated as an exploratory estimate for the global mean and PEOU (Section III-B4a). Psychometric robustness was confirmed ($\alpha_{\text{global}}=0.866$; $\alpha_{\text{PU}}=0.867$; $\alpha_{\text{PEOU}}=0.898$; all above 0.80 [47]). PU achieved $\bar{x}_{\text{PU}}=4.55$, with PTAM002 ($\bar{x}=4.76$; 84% at level 5) as the highest-valued item. The voice navigation initial learning curve (PTAM007: $\bar{x}=3.56$; $SD=1.36$) and complete functional mastery (PTAM012: $\bar{x}=3.48$) are the two priority improvement dimensions of the PEOU construct ($\bar{x}_{\text{PEOU}}=3.85$).

The **third conclusion** concerns the efficacy of the TDDM4IoTTS iterative cycle for systematic accessibility improvement (RQ3=confirmed). The methodology [22] with accessibility as a verifiable acceptance criterion from Phase 5 produced a global 64.0% reduction in WCAG 2.1 warnings (75 to 27), with total elimination on camera and photo screens (100%) and reductions $>70\%$ on settings (77.8%) and main screens (81.8%). The residual limitation on the messaging screen (38.1%) represents a traceable technical debt with a defined resolution protocol.

The **fourth conclusion** concerns the resolution of technological fragmentation documented in the literature. The AI

components for accessibility—neural TTS, transformer-based STT, image captioning, and OCR—which the literature has developed in isolation [19], [20], [37]–[39] are integrable in a coherent, stable, privacy-guaranteed mobile messaging architecture on low-cost institutional hybrid infrastructure.

Five high-priority future research directions follow: (i) optimising TTS latency via non-autoregressive architectures (Fast-Speech2, VITS) or predictive voice-segment caching to reach the 2 s conversational threshold [19]; (ii) designing an accessible onboarding flow with personalised voice command calibration to the user's disfluency profile, following the adaptive training paradigm of Costanzo *et al.* [12]; (iii) integrating sign language recognition as an additional input/output modality, taking the LSTM-GRU models of Kothadiya *et al.* [14] and the multilingual systems of Saleem *et al.* [13] as references; (iv) architectural refactoring of the messaging screen with dynamic unique accessibility identifiers per list element to eliminate the 13 residual warnings and raise global system satisfaction (PTAM012: $\bar{x}=3.48$); (v) expanding the study to samples stratified by disability type, age range, and mobile connectivity conditions (4G/LTE low-bandwidth networks) to increase results generalisability and identify technology acceptance moderation factors specific to each sensory profile.

ACKNOWLEDGMENT

The authors thank the Universidad Técnica Estatal de Quevedo (UTEQ) for its technical, financial, and institutional support. They also express their gratitude to the 25 study participants, whose collaboration was fundamental to the empirical validation of UniVoice.

REFERENCES

- [1] World Health Organization and World Bank, *World Report on Disability*. Geneva, Switzerland: WHO Press, 2011. [Online]. Available: <https://www.who.int/publications/i/item/9789241564182>
- [2] World Health Organization, "World report on hearing," WHO Press, Geneva, Switzerland, Tech. Rep., 2021. [Online]. Available: <https://www.who.int/publications/i/item/9789240020481>
- [3] —, "Blindness and vision impairment — fact sheet," WHO Media Centre, 2023, accessed 2024. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment>
- [4] National Institute on Deafness and Other Communication Disorders, "Statistics on voice, speech, and language," NIDCD, U.S. Department of Health & Human Services, 2024, accessed 2024. [Online]. Available: <https://www.nidcd.nih.gov/health/statistics/statistics-voice-speech-and-language>
- [5] American Speech-Language-Hearing Association, "Communication disorders: Prevalence and incidence data," ASHA, 2023, accessed 2024. [Online]. Available: <https://www.asha.org/policy/>
- [6] Pan American Health Organization, "Hearing health in the region of the americas," PAHO/WHO, 2023, accessed 2024. [Online]. Available: <https://www.paho.org/en/topics/hearing-health>
- [7] Consejo Nacional para la Igualdad de Discapacidades, "Tabla de registros de personas con discapacidad por género: discapacidades auditivas, visuales y de lenguaje," Base de datos institucional CONADIS, 2024, consultado en 2024. [Online]. Available: <https://www.consejodiscapacidades.gob.ec/>
- [8] V. Bercaru and N. Popescu, "A systematic review of accessibility techniques for online platforms: Current trends and challenges," *Applied Sciences*, vol. 14, no. 22, p. 10337, 2024.
- [9] P. Weichbroth, "Usability of mobile applications: A systematic literature study," *IEEE Access*, vol. 8, pp. 55 563–55 577, 2020.
- [10] Y. Elsahar, S. Hu, K. Bouazza-Marouf, D. Kerr, and A. Mansor, "Augmentative and alternative communication (AAC) advances: A review of configurations for individuals with a speech disability," *Sensors*, vol. 19, no. 8, p. 1911, 2019.

- [11] A. Bleakley, D. Rough, A. Roper, S. Lindsay, M. Porcheron, M. Lee, S. A. Nicholson, B. R. Cowan, and L. Clark, "Exploring smart speaker user experience for people who stammer," in *Proceedings of the 24th International ACM SIGACCESS Conference on Computers and Accessibility (ASSETS)*. ACM, 2022, pp. 1–17.
- [12] F. Costanzo, E. Fucà, C. Caciolo, D. Ruà, S. Smolley, D. Weissberg, and S. Vicari, "Talkitt: Toward a new instrument based on artificial intelligence for augmentative and alternative communication in children with Down syndrome," *Frontiers in Psychology*, vol. 14, p. 1176683, 2023.
- [13] M. I. Saleem, A. Siddiqui, S. Noor, M.-A. Luque-Nieto, and P. Otero, "A novel machine learning based two-way communication system for deaf and mute," *Applied Sciences*, vol. 13, no. 1, p. 453, 2023.
- [14] D. Kothadiya, C. Bhatt, K. Sapariya, K. Patel, A.-B. Gil-González, and J. M. Corchado, "DeepSign: Sign language detection and recognition using deep learning," *Electronics*, vol. 11, no. 11, p. 1780, 2022.
- [15] M. F. Story, "Maximizing usability: The principles of universal design," in *Assistive Technology*, 1998, vol. 10, no. 1, pp. 4–12.
- [16] World Wide Web Consortium, "Web content accessibility guidelines (WCAG) 2.1," W3C, Tech. Rep., 2018, w3C Recommendation. [Online]. Available: <https://www.w3.org/TR/WCAG21/>
- [17] S. Chen, C. Chen, L. Fan, M. Fan, X. Zhan, and Y. Liu, "Accessible or not? an empirical investigation of android app accessibility," *IEEE Transactions on Software Engineering*, vol. 48, no. 10, pp. 3954–3968, 2022.
- [18] P. Acosta-Vargas, J. Gueña-Moya, J. Jadán-Guerrero, C. Alvites-Huamani, and L. Salvador-Ullauri, "Towards accessibility assessment with a combined approach for native mobile applications," in *Lecture Notes in Networks and Systems*, vol. 265. Springer, 2021, pp. 234–241.
- [19] X. Tan, T. Qin, F. Soong, and T. Liu, "A survey on neural speech synthesis," *arXiv preprint*, 2021, arXiv:2106.15561.
- [20] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A framework for self-supervised learning of speech representations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 12 449–12 460.
- [21] P. Dognin, I. Melnyk, Y. Mroueh, I. Padhi, M. Rigotti, J. Ross, Y. Schiff, R. A. Young, and B. Belgodere, "Image captioning as an assistive technology: Lessons learned from vizwiz 2020 challenge," *Journal of Artificial Intelligence Research*, vol. 73, pp. 437–459, 2021.
- [22] G. Guerrero-Ulloa, M. J. Hornos, and C. Rodríguez-Domínguez, "TDDM4IoT: A test-driven development methodology for Internet of Things (IoT)-based systems," in *Communications in Computer and Information Science*, vol. 1193. Springer, 2020, pp. 41–55.
- [23] C. F. da Silva, S. B. L. Ferreira, and J. F. M. Ramos, "WhatsApp accessibility from the perspective of visually impaired people," in *Proceedings of the 15th Brazilian Symposium on Human Factors in Computing Systems*. ACM, 2016, pp. 1–10.
- [24] W. Zahida, V. Effendy, and A. Hadikusuma, "Improving messenger accessibility for elderly users using user centered design (UCD) methods (study case: WhatsApp)," in *2021 International Conference on Software Engineering & Computer Systems and 4th International Conference on Computational Science and Information Management (ICSECS-ICCSIM)*. IEEE, 2021, pp. 273–278.
- [25] M. A. Swati, T. M. Madni, U. I. Janjua, and I. Ahmad, "Accessibility of social media application for blinds," in *2021 4th International Conference on Computing and Information Sciences (ICCIS)*. IEEE, 2021, pp. 1–6.
- [26] R. Rumini and A. I. Astrid, "Evaluating the usability of heuristics within Telegram using the linear regression method," in *2019 4th International Conference on Information Technology, Information Systems and Electrical Engineering (ICITISEE)*. IEEE, 2019, pp. 469–473.
- [27] H. Kim, H. Hwang, S. Gwak, J. Yoon, and K. Park, "Improving communication and promoting social inclusion for hearing-impaired users: Usability evaluation and design recommendations for assistive mobile applications," *PLOS ONE*, vol. 19, no. 7, p. e0305726, 2024.
- [28] M. Ballantyne, A. Jha, A. Jacobsen, J. S. Hawker, and Y. N. El-Ghaly, "Study of accessibility guidelines of mobile applications," in *Proceedings of the 17th International Conference on Mobile and Ubiquitous Multimedia*. ACM, 2018, pp. 305–315.
- [29] M. Huang, G. McGrew, and C. Bodine, "Social media apps: A paradigm for examining usability of mobile apps for working-age adults with mild-moderate cognitive disabilities," *ACM Transactions on Accessible Computing*, vol. 18, no. 1, pp. 1–38, 2025.
- [30] G. Guerrero-Ulloa, M. J. Hornos, and C. Rodríguez-Domínguez, "BellChat: Una aplicación web inclusiva para personas con discapacidades visuales y auditivas," in *Proceedings of the XVI Multi-Conference on Computer Science and Information Systems*, 2021, ISBN: 978-989-758-534-0.
- [31] A. Yeratziotis, A. Achilleos, S. Koumou, G. Zampas, R. A. Thibodeau, G. Geratziotis, G. A. Papadopoulos, I. Iasonas, and C. Kronis, "Making social media applications inclusive for deaf end-users with access to sign language," *Multimedia Tools and Applications*, vol. 82, no. 29, pp. 46 185–46 215, 2023.
- [32] I. Chang, J. Castillo, and H. Montes, "Technology-based social innovation: Smart city inclusive system for hearing impairment and visual disability citizens," *Sensors*, vol. 22, no. 3, p. 848, 2022.
- [33] N. Janaka, J. Gao, L. Zhu, S. Zhao, L. Lyu, P. Xu, M. Nabokow, S. Wang, and Y. Ong, "GlassMessaging: Towards ubiquitous messaging using OHMDs," *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 7, no. 3, p. 100, 2023.
- [34] M. Mateos-Sanchez, A. Casado Melo, L. Sánchez Blanco, and A. M. Feroso García, "Chatbot, as educational and inclusive tool for people with intellectual disabilities," *Sustainability*, vol. 14, no. 3, p. 1520, 2022.
- [35] K. Chemnad and A. Othman, "Digital accessibility in the era of artificial intelligence—bibliometric analysis and systematic review," *Frontiers in Artificial Intelligence*, vol. 7, p. 1349668, 2024.
- [36] S. Ji, J. Zuo, M. Fang, Z. Jiang, F. Chen, X. Duan, B. Huai, and Z. Zhao, "TextrolSpeech: A text style control speech corpus with codec language text-to-speech models," in *ICASSP 2024 – 2024 IEEE International Conference on Acoustics, Speech and Signal Processing*. IEEE, 2024, pp. 10 301–10 305.
- [37] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, vol. 202. PMLR, 2023, pp. 28 492–28 518.
- [38] Y. Ming, N. Hu, C. Fan, F. Feng, J. Zhou, and H. Yu, "Visuals to text: A comprehensive review on automatic image captioning," *IEEE/CAA Journal of Automatica Sinica*, vol. 9, no. 8, pp. 1339–1365, 2022.
- [39] S. Mukherjee, H. Tyagi, P. Tyagi, N. Singh, and S. Bhardwaj, "OCR using Python and its application," *Journal of Advanced Zoology*, vol. 44, no. S-3, pp. 1083–1092, 2023.
- [40] M. Papatsimouli, P. Sarigiannidis, and G. F. Fragulis, "A survey of advancements in real-time sign language translators: Integration with IoT technology," *Technologies*, vol. 11, no. 4, p. 83, 2023.
- [41] R. Hernández Sampieri, C. Fernández Collado, and P. Baptista Lucio, *Metodología de la Investigación*, 6th ed. Mexico City, Mexico: McGraw-Hill Education, 2014.
- [42] K. M. Safiya and R. Pandian, "A real-time image captioning framework using computer vision to help the visually impaired," *Multimedia Tools and Applications*, vol. 83, pp. 59 413–59 438, 2024.
- [43] A. R. Lahitani, A. E. Permanasari, and N. A. Setiawan, "Cosine similarity to determine similarity measure: Study case in online essay assessment," in *2016 4th International Conference on Cyber and IT Service Management*. IEEE, 2016, pp. 1–6.
- [44] A. Shostack, *Threat Modeling: Designing for Security*. Indianapolis, IN: John Wiley & Sons, 2014.
- [45] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," *MIS Quarterly*, vol. 13, no. 3, pp. 319–340, 1989.
- [46] R. J. Holden and B.-T. Karsh, "The technology acceptance model: Its past and its future in health care," *Journal of Biomedical Informatics*, vol. 43, no. 1, pp. 159–172, 2010.
- [47] J. C. Nunnally, *Psychometric Theory*, 2nd ed. New York, NY, USA: McGraw-Hill, 1978.
- [48] F. Faul, E. Erdfelder, A.-G. Lang, and A. Buchner, *G*Power 3: A flexible statistical power analysis program for the social, behavioral, and biomedical sciences*, 2007.
- [49] J. Cohen, *Statistical Power Analysis for the Behavioral Sciences*, 2nd ed. Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.
- [50] World Medical Association, "Declaration of Helsinki: Ethical principles for medical research involving human subjects," *JAMA*, vol. 310, no. 20, pp. 2191–2194, 2013.
- [51] M. P. Galindo-Villardón, "An alternative for simultaneous representation: HJ-Biplot," *Questió: Quaderns d'Estadística, Sistemes, Informàtica i Investigació Operativa*, vol. 10, no. 1, pp. 13–23, 1986. [Online]. Available: <https://eudml.org/doc/40045>
- [52] M. Cubilla-Montilla, A. B. Nieto-Librero, M. P. Galindo-Villardón, and C. A. Torres-Cubilla, "Sparse HJ biplot: A new methodology via elastic net," *Mathematics*, vol. 9, no. 11, p. 1298, 2021.
- [53] F. H. F. Botelho, "Accessibility to digital technology: Virtual barriers, real opportunities," *Assistive Technology*, vol. 33, no. sup1, pp. 27–34, 2021.

- [54] T. Stefanov, S. Varbanova, M. Stefanova, and I. Alinski, "Mobile app prototype supporting people with special needs," *TEM Journal*, pp. 1871–1880, 2024.
- [55] J. Ara, C. Sik-Lanyi, and A. Kelemen, "Accessibility engineering in web evaluation process: A systematic literature review," *Universal Access in the Information Society*, vol. 23, no. 2, pp. 653–686, 2024.



Gleiston Guerrero Ulloa received the Ph.D. degree in Computer Science from the Universidad de Granada, Granada, Spain, in 2021. He is currently an Associate Professor at the Faculty of Computing and Digital Design, Universidad Técnica Estatal de Quevedo (UTEQ), Quevedo, Ecuador. His research interests include software engineering, IoT-based system development, test-driven development methodologies, accessible and inclusive computing, and human-machine interaction. He is the co-developer of the TDDM4IoTS methodology and

author of BellChat, the first inclusive web messaging application integrating TTS and STT for users with visual and hearing disabilities.



Jhon Byron Leturne Pluas received the B.Sc. degree in Software Engineering from the Universidad Técnica Estatal de Quevedo (UTEQ), Quevedo, Ecuador, in 2025. He is currently a Graduate Researcher at the Faculty of Computing Science, UTEQ. His research interests include inclusive mobile application development, AI-assisted accessibility systems, human-machine interaction for people with sensory disabilities, and test-driven development methodologies for IoT-based systems.